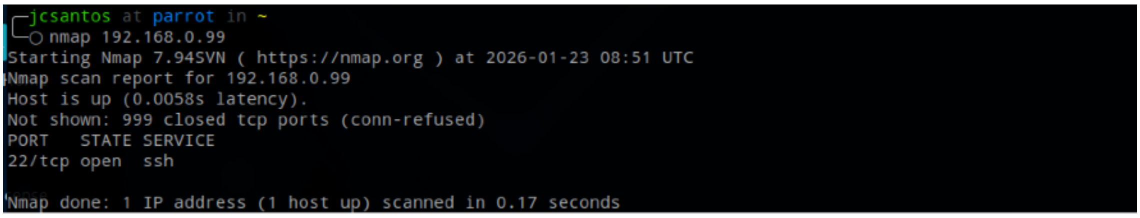
Ejercicios de Nmap (Redes y Seguridad Informática)

📘 1. Comprobar conectividad básica y escaneo simple

nmap 192.168.1.143

- Escanea el host con IP 192.168.1.143 para detectar puertos abiertos comunes.



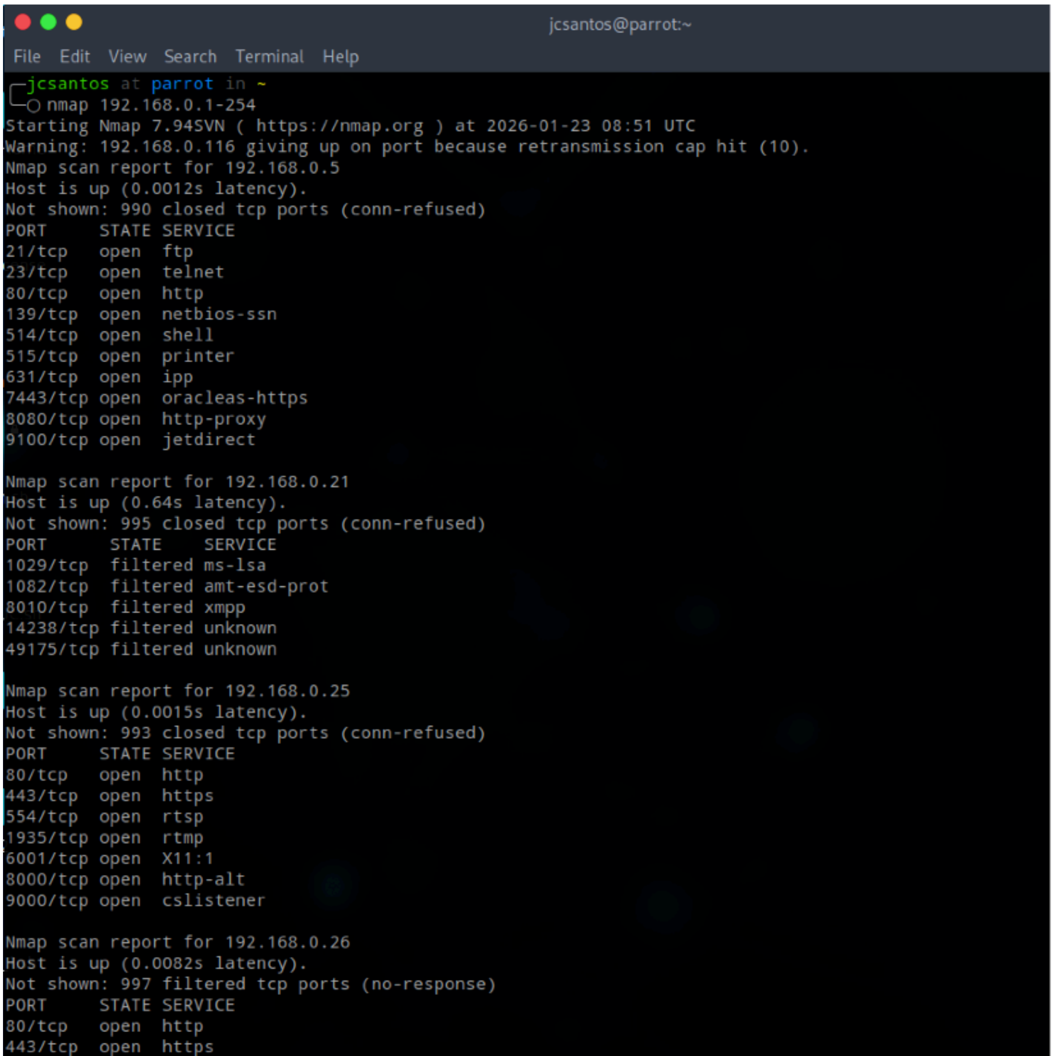
```
┌─jcsantos at parrot in ~
└─○ nmap 192.168.0.99
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 08:51 UTC
Nmap scan report for 192.168.0.99
Host is up (0.0058s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds
```

📘 2. Escaneo de un rango de IP

nmap 192.168.1.1-254

- Escanea todas las direcciones IP del rango 192.168.1.1 a 192.168.1.254.



```
jcsantos@parrot:~
File  Edit  View  Search  Terminal  Help
┌─jcsantos at parrot in ~
└─○ nmap 192.168.0.1-254
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 08:51 UTC
Warning: 192.168.0.116 giving up on port because retransmission cap hit (10).
Nmap scan report for 192.168.0.5
Host is up (0.0012s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
23/tcp   open  telnet
80/tcp   open  http
139/tcp  open  netbios-ssn
514/tcp  open  shell
515/tcp  open  printer
631/tcp  open  ipp
7443/tcp open  oracleas-https
8080/tcp open  http-proxy
9100/tcp open  jetdirect

Nmap scan report for 192.168.0.21
Host is up (0.64s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE    SERVICE
1029/tcp  filtered ms-lsa
1082/tcp  filtered amt-esd-prot
8010/tcp  filtered xmpp
14238/tcp filtered unknown
49175/tcp filtered unknown

Nmap scan report for 192.168.0.25
Host is up (0.0015s latency).
Not shown: 993 closed tcp ports (conn-refused)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
554/tcp  open  rtsp
1935/tcp open  rtmp
6001/tcp open  X11:1
8000/tcp open  http-alt
9000/tcp open  cslistener

Nmap scan report for 192.168.0.26
Host is up (0.0082s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
```

📘 3. Escaneo de varias direcciones IP específicas

nmap 192.168.1.1 192.168.1.4 192.168.1.43

- Escanea múltiples IP individuales especificadas manualmente.

```
┌─jcsantos at parrot in ~
└─○ nmap 192.168.0.78 192.168.0.90 192.168.0.99
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 09:03 UTC
Nmap scan report for 192.168.0.78
Host is up (0.00064s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh

Nmap scan report for 192.168.0.90
Host is up (0.030s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT    STATE SERVICE
22/tcp  open  ssh
80/tcp  open  http
443/tcp open  https

Nmap scan report for 192.168.0.99
Host is up (0.0010s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh

Nmap done: 3 IP addresses (3 hosts up) scanned in 0.54 seconds
```

📘 4. Escaneo de puertos específicos

nmap -p 1042 192.168.1.143

- Escanea solo el puerto 1042 en la IP dada.

```
┌─jcsantos at parrot in ~
└─○ nmap -p 22 192.168.0.1-254
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 09:05 UT
Nmap scan report for 192.168.0.5
Host is up (0.0011s latency).

PORT   STATE  SERVICE
22/tcp closed ssh

Nmap scan report for 192.168.0.25
Host is up (0.0013s latency).

PORT   STATE  SERVICE
22/tcp closed ssh

Nmap scan report for 192.168.0.26
Host is up (0.0088s latency).

PORT   STATE    SERVICE
22/tcp filtered ssh

Nmap scan report for 192.168.0.27
Host is up (0.046s latency).

PORT   STATE    SERVICE
22/tcp filtered ssh

Nmap scan report for 192.168.0.28
Host is up (0.11s latency).

PORT   STATE  SERVICE
22/tcp closed ssh

Nmap scan report for 192.168.0.40
Host is up (0.045s latency).

PORT   STATE  SERVICE
22/tcp closed ssh

Nmap scan report for 192.168.0.47
Host is up (0.054s latency).
```

5. Escaneo de puertos UDP

nmap -sU -p 7 11 15 18 19 20 21 22 51 143 514 8080 192.168.1.143

- Escaneo en modo UDP de puertos específicos.
- El parámetro -sU indica que se trata de un escaneo de puertos UDP.

```
┌─jcsantos at parrot in ~
└─○ sudo nmap -sU -p 7 11 15 18 19 20 21 22 51 143 514 8080 192.168.1.1-254
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 09:09 UTC
Nmap done: 265 IP addresses (0 hosts up) scanned in 20.87 seconds
┌─jcsantos at parrot in ~
└─○
```

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6. Escaneo del sistema operativo

nmap -O -osscan-guess localhost

- Detecta el sistema operativo del host local.
- El parámetro --osscan-guess permite adivinar el sistema si no se reconoce con certeza.

```
┌─jcsantos at parrot in ~
└─○ sudo nmap -O -osscan-guess 192.168.0.209
[sudo] password for jcsantos:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 10:19 UTC
Nmap scan report for 192.168.0.209
Host is up (0.000068s latency).
Not shown: 998 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
2022/tcp open  down
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.44 seconds
┌─jcsantos at parrot in ~
└─○
```

7. Mostrar versión de los servicios

nmap -sV -version-all localhost

- Muestra la versión de los servicios detectados en el host local.

```
┌─jcsantos at parrot in ~
└─○ sudo nmap -sV -version-all 127.0.0.1
[sudo] password for jcsantos:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 09:13 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 998 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
22/tcp   open  ssh     (protocol 2.0)
2022/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u7 (protocol 2.0)
1 service unrecognized despite returning data. If you know the service/version, please submit the fo
SF-Port22-TCP:V=7.94SVN%I=9%D=1/23%Time=69733BC6%P=x86_64-pc-linux-gnu%r(N
SF:ULL,25,"SSH-2\.0-tinyssh_20230101-1\x20xaoqJKlB\r\n")%r(Hello,25,"SSH-2
SF:\.0-tinyssh_20230101-1\x20gAZMjayZ\r\n")%r(Help,25,"SSH-2\.0-tinyssh_20
SF:230101-1\x20MuDGw3cW\r\n")%r(LPDString,25,"SSH-2\.0-tinyssh_20230101-1\
```

📘 8. Mostrar información del sistema Nmap instalado

nmap -SV --version-1ll localhost

- Posible errata tipográfica, probablemente se refiere a:

nmap -sV --version-intensity 9 localhost

```
┌─jcsantos at parrot in ~
└─○ nmap -SV --version || 127.0.0.1
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.4 openssl-3.0.18 libssh2-1.10.0 libz-1.2.13 libpcre2-10.42 libpcap-1.10.3
nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
┌─jcsantos at parrot in ~
```

📘 9. Ver interfaces de red

nmap --iflist localhost

- Muestra todas las interfaces de red y rutas de la máquina local.

```
┌─jcsantos at parrot in ~
└─○ nmap --iflist localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 09:20 UTC
************************INTERFACES************************
DEV   (SHORT) IP/MASK                     TYPE     UP MTU   MAC
lo    (lo)    127.0.0.1/8                 loopback up 65536
lo    (lo)    ::1/128                     loopback up 65536
ens33 (ens33) 192.168.0.209/23            ethernet up 1500  00:0C:29:13:85:B3
ens33 (ens33) fe80::183a:4430:ba45:6098/64 ethernet up 1500  00:0C:29:13:85:B3

**************************ROUTES*************************
DST/MASK                      DEV   METRIC GATEWAY
192.168.0.0/23                ens33 100
0.0.0.0/0                     ens33 100    192.168.0.1
::1/128                       lo    0
fe80::183a:4430:ba45:6098/128 ens33 0
fe80::/64                     ens33 1024
ff00::/8                      ens33 256
```

📘 10. Enviar paquetes TCP ACK

nmap -sA 192.168.1.143

```
┌─jcsantos at parrot in ~
└─○ sudo nmap -sA 192.168.0.81
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 09:24 UTC
Nmap scan report for 192.168.0.81
Host is up (0.00064s latency).
All 1000 scanned ports on 192.168.0.81 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 24:BE:05:21:18:3B (Hewlett Packard)

Nmap done: 1 IP address (1 host up) scanned in 21.36 seconds
┌─jcsantos at parrot in ~
```

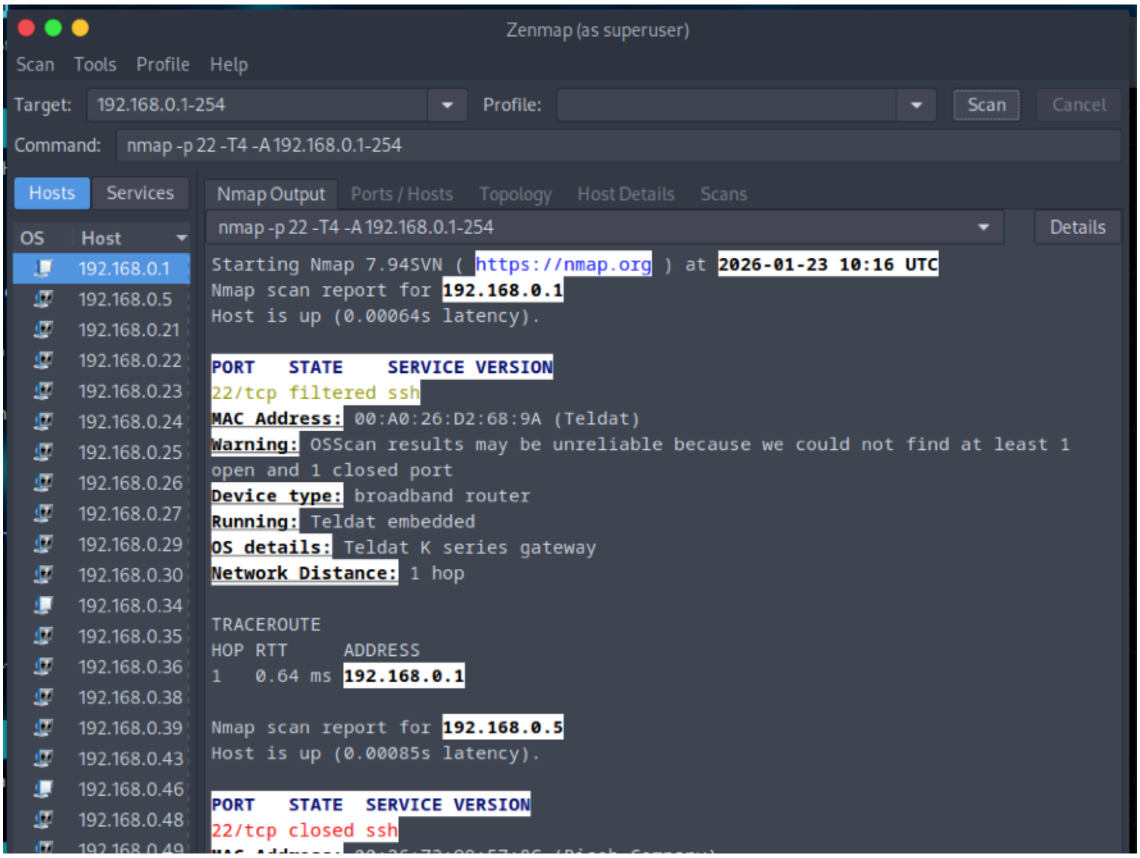
- Escanea con paquetes TCP ACK para detectar hosts activos incluso tras firewall.

📘 11. Actividad propuesta 1.9: Zenmap GUI para Nmap

🧩Instrucciones del ejercicio:

Instala **Zenmap** (entorno gráfico para Nmap) en tu máquina Windows.

Investiga y explica **la utilidad de las diferentes pestañas** que ofrece la aplicación.



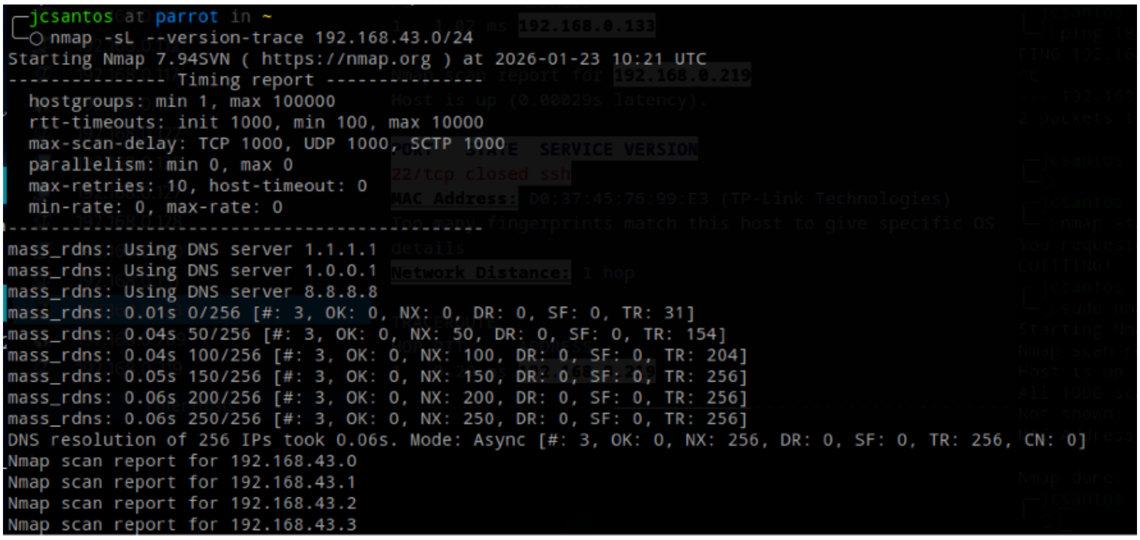
📘 12. Otros comandos avanzados (mencionados en la imagen):

- Detección de DNS inverso:

nmap -sL 192.168.43.0/24

- Escaneo de versión con --version-trace
- Exclusión de host:

nmap 192.168.1.0/24 --exclude 192.168.1.1



```
┌─jcsantos at parrot in ~
└─○ nmap -sL --version-trace 192.168.43.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-23 10:21 UTC
--------------- Timing report ---------------
  hostgroups: min 1, max 100000
  rtt-timeouts: init 1000, min 100, max 10000
  max-scan-delay: TCP 1000, UDP 1000, SCTP 1000
  parallelism: min 0, max 0
  max-retries: 10, host-timeout: 0
  min-rate: 0, max-rate: 0
---------------------------------------------
mass_rdns: Using DNS server 1.1.1.1
mass_rdns: Using DNS server 1.0.0.1
mass_rdns: Using DNS server 8.8.8.8
mass_rdns: 0.01s 0/256 [#: 3, OK: 0, NX: 0, DR: 0, SF: 0, TR: 31]
mass_rdns: 0.04s 50/256 [#: 3, OK: 0, NX: 50, DR: 0, SF: 0, TR: 154]
mass_rdns: 0.04s 100/256 [#: 3, OK: 0, NX: 100, DR: 0, SF: 0, TR: 204]
mass_rdns: 0.05s 150/256 [#: 3, OK: 0, NX: 150, DR: 0, SF: 0, TR: 256]
mass_rdns: 0.06s 200/256 [#: 3, OK: 0, NX: 200, DR: 0, SF: 0, TR: 256]
mass_rdns: 0.06s 250/256 [#: 3, OK: 0, NX: 250, DR: 0, SF: 0, TR: 256]
DNS resolution of 256 IPs took 0.06s. Mode: Async [#: 3, OK: 0, NX: 256, DR: 0, SF: 0, TR: 256, CN: 0]
Nmap scan report for 192.168.43.0
Nmap scan report for 192.168.43.1
Nmap scan report for 192.168.43.2
Nmap scan report for 192.168.43.3
```